

Scope document

Referrals and Appointments

9 July, 2026

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1 Document Control

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2 Executive Summary

The Project - NHS England has invested significant time and expertise into developing a proof of concept (PoC) for the BaRS Endpoint Catalogue Service (EPCSe), recognising its importance as a foundational capability within the broader BaRS ecosystem. Building on the success of this PoC, NHSE has now commissioned Accenture to deliver the EPCSe solution, ensuring that it adheres to established NHSE development standards and aligns with tooling guidance set out in the NHS [Tech Radar](#)¹.

Accenture has mobilised a dedicated, multidisciplinary team to design, build, and deliver the EPCSe in accordance with the defined requirements. This team will work in close partnership with NHSE subject matter experts who shaped the PoC, ensuring continuity of understanding, efficient decision making, and alignment with strategic objectives. The delivery approach will be structured to maintain momentum, safeguard quality, and ensure that the solution is developed, tested, and deployed in line with agreed expectations and timelines.

The Need - The EPCSe is a critical enabler for BaRS, underpinning the integrity, reliability, and security of endpoint information used across NHS services. As BaRS adoption accelerates, the accuracy and governance of endpoint data will become increasingly important—not only for ensuring correct message routing but also for protecting operational resilience and clinical safety. Currently, endpoint management is labour intensive, distributed across multiple systems, and reliant on manual processes that introduce risk, delay, and the potential for inconsistencies. As the scale and adoption of BaRS grows, this model is no longer sustainable. The EPCSe addresses these challenges by providing a centralised, robust, and policy aligned mechanism for managing endpoint information, ensuring it remains accurate, secure, and accessible to authorised users. The platform will also ensure that appropriate technologies and interfaces are used when interacting with endpoint data, enabling a more seamless and auditable end-to-end process.

What Success Looks Like - Success will be defined by the delivery of a functional, compliant, and scalable Endpoint Catalogue (EPC) that supports national BaRS integration and operational use. Accenture will be responsible for liaising with the concerned teams for integrating EPC with the BaRS Proxy, perform robust data migration, end to end testing, and live service deployment. The delivery approach will follow Agile principles, enabling iterative progress, transparent reporting, early risk surfacing, and regular stakeholder validation. Each milestone will be supported by clear acceptance criteria, dependency tracking, and alignment with governance structures. The delivery team will also produce a comprehensive suite of supporting documentation to ensure longterm value and operational readiness. This will include a fully scoped backlog, technical architecture artefacts, test plans and results. Knowledge transfer will cover EPC architecture, update and maintenance mechanisms. This will equip NHS teams with the capability and confidence to manage and extend the EPC once transition into live service is complete.

1. <https://radar.engineering.england.nhs.uk/view-radar/index.html>

3 Background & Context

The Booking and Referral Standard (BaRS) is a nationally mandated interoperability standard designed to enable the consistent, secure, and timely exchange of booking and referral information across NHS service providers. By ensuring that patient information is transmitted in a structured, clinically meaningful format, BaRS supports operational efficiency and enhances clinical decision making. The standard relies on an API Proxy to route messages between sending and receiving supplier systems, ensuring they reach the correct destination safely and reliably.

The Endpoint Catalogue Service (EPCSe) is a critical enabler within this ecosystem. It provides the centralised capability required to look up, validate, and use accurate digital endpoints for a wide range of Healthcare Services. This includes BaRS endpoints, as well as those used by other national interoperability initiatives such as GP Connect and ITK in the future. By linking business identifiers to the appropriate physical endpoints, the EPCSe introduces a more flexible, durable, and resilient model for managing the systems that support NHS services.

At present, endpoint management from a BaRS perspective is largely manual and operationally burdensome. The Directory of Services (DoS) endpoint processes are jointly managed by BaRS and the UEC DoS team, requiring multiple systems, manual intervention, and coordination across teams to ensure messages are routed correctly. This approach is prone to delays, errors, and inconsistency—particularly when new endpoints are introduced or existing ones are updated or retired. While BaRS currently manages endpoints for pharmacy services and selected other services—including the Northwest NHS 111 service—the growing scale of BaRS adoption means that the existing model is increasingly unsustainable. Without modernisation, it will introduce operational inefficiencies and create avoidable clinical, technical, and business risks.

To mitigate these challenges and support future scaling, a more robust and self-managed solution is essential. Developing a production grade Endpoint Catalogue Service will streamline and standardise endpoint management at the point new services join BaRS, reducing the operational overhead and enabling updates to be executed reliably by a single operator using one integrated system. This centralised and automated model will improve governance, significantly reduce the risk of routing errors, and ensure the NHS maintains a high quality, up to date set of endpoints in a live environment.

Strategic Alignment

The introduction of the EPCSe aligns directly with NHS England's broader strategic priorities around interoperability, data quality, digital operational resilience, and the safe scaling of national services. By modernising and centralising endpoint management, the EPCSe supports the NHS ambition of creating repeatable, sustainable digital foundations that enable seamless, standards based communication across care settings.

Current State vs. Desired Future State

Current State: A proof of concept has been created to automate elements of BaRS endpoint management. However, the PoC still relies on two people and multiple systems to maintain accuracy and manage updates. This limits operational efficiency and restricts the NHS's ability to scale BaRS safely and quickly.

Future State: A streamlined endpoint management capability for BaRS. This target model will enable endpoint updates to be completed quickly, confidently with support from the BaRS team, using one unified system in an integration and live production environment. This will reduce operational risk, enhance service resilience, and support rapid national uptake. The future state is documented in more detail in this requirements document: [Endpoint Catalogue Service Requirements V1.8 2025-02-12 \(New CLEAN\) Draft.pdf](#)²

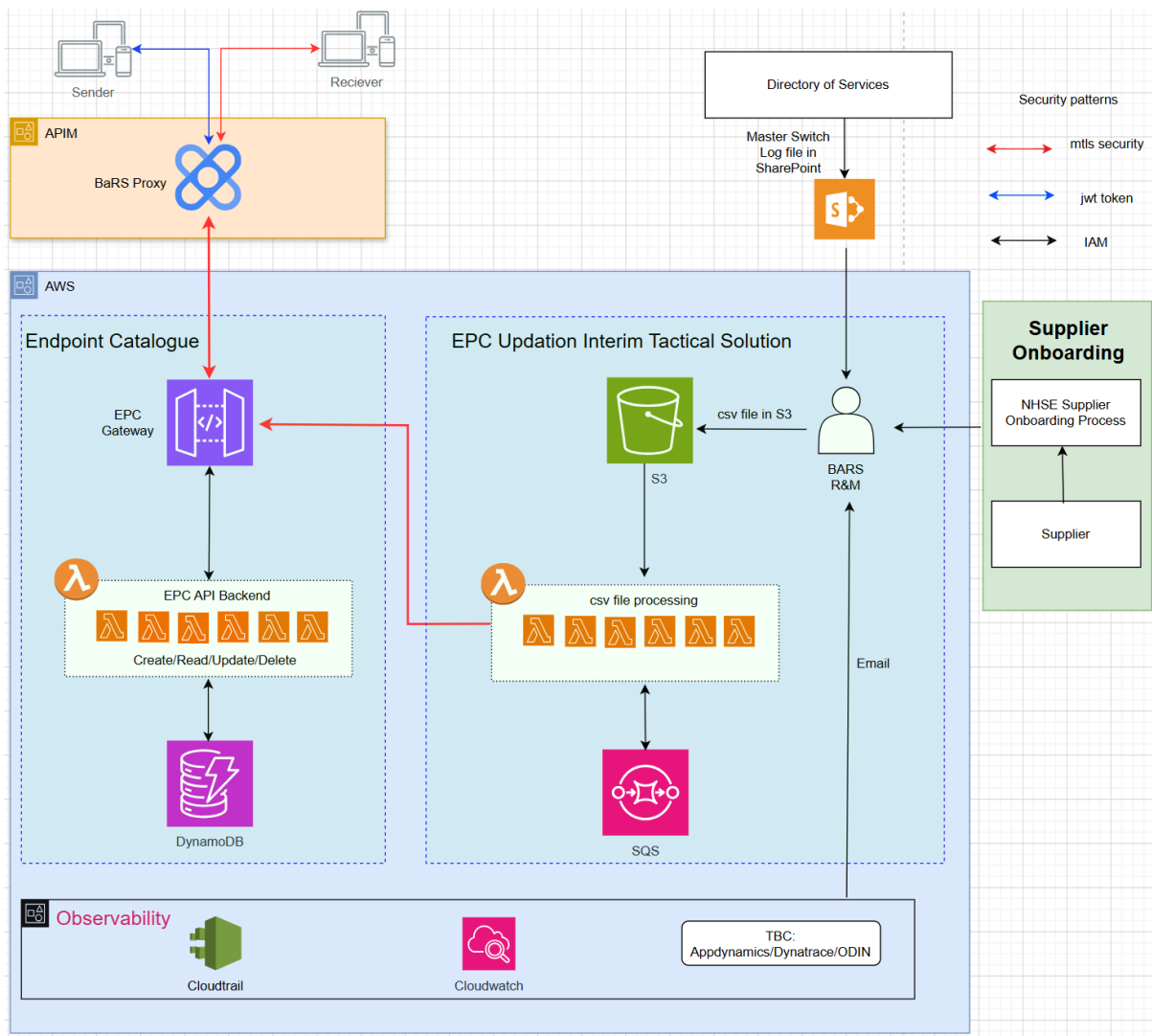
2. <https://nhsd-confluence.digital.nhs.uk/download/attachments/1285824056/>

Endpoint%20Catalogue%20Service%20Requirements%20V1.8%202025-02-12%20%28New%20CLEAN%29%20D
raft.pdf?api=v2&modificationDate=1773660193334&version=1

4 Scope

4.1 Architecture overview

The solution is implemented as a serverless, event-driven Endpoint Catalogue platform within the BaRS AWS workload account. It provides two primary capabilities: 1) a real-time API for querying and managing endpoint data, and 2) a batch ingestion pipeline for processing CSV files. In future, additions/updates may be singular or bulk and need to support national and proprietary directory of services (the format for ingestion will remain consistent for all). External consumers interact with the platform through API Management (APIM), which provides a controlled entry point for security, routing, and policy enforcement. Behind this, managed AWS services handle compute, messaging, storage, and persistence, allowing the platform to scale automatically while minimising operational overhead. The architecture separates real-time request processing from bulk data ingestion, ensuring that large data updates do not impact live API performance or availability.



4.2 Functional Overview

Below, are high level descriptions of the functional use cases for the BaRS Endpoint Catalogue. These will be detailed at a lower level in process flows and designs separately.

The scope of this document reflects what is being built under SOW13. This will be expanded as part of future phases. More details on future scope can be found here: [Endpoint Catalogue Service Requirements V1.8 2025-02-12 \(New CLEAN\) Draft.pdf](https://nhsd-confluence.digital.nhs.uk/download/attachments/1285824056/Endpoint%20Catalogue%20Service%20Requirements%20V1.8%202025-02-12%20%28New%20CLEAN%29%20Draft.pdf?api=v2&modificationDate=1773660193334&version=1)³

3. <https://nhsd-confluence.digital.nhs.uk/download/attachments/1285824056/Endpoint%20Catalogue%20Service%20Requirements%20V1.8%202025-02-12%20%28New%20CLEAN%29%20Draft.pdf?api=v2&modificationDate=1773660193334&version=1>

4.2.1 Real-time API request flow

1. Sender initiates a request to the receiver
2. Request enters APIM
3. The BaRS Proxy on APIM routes the request to API Gateway
4. API Gateway invokes API Backend (Lambda functions)
5. Backend performs core business logic
6. Backend reads Endpoint information from DynamoDB
7. Backend relays Endpoint information via API Gateway to the BaRS Proxy
8. BaRS proxy forwards the Sender's request to the receiver

4.2.2 Endpoint management (Add, Update, Delete)

Broadly, our use cases for endpoint management can be categories as adding, updating or deleting an endpoint / endpoint template.

4.2.2.1 Supplier Onboarding – Add endpoint workflow

1. Supplier completes onboarding formalities
2. Supplier specifies if they want to create an endpoint or an endpoint template
3. Based on the specification, the NHSE BaRS team creates a CSV file and provides it to the BaRS R&M team
4. The BaRS R&M team uploads the CSV file to S3 bucket
5. File upload triggers the automated validation process by the “file ingestion” Lambda which does a validation to ensure presence of correct file format and mandatory fields
6. The next step involves the EPC updating lambda which processes every row of the CSV file and sends details to API backend
7. The API backend adds the endpoint(s) to the catalogue in DynamoDB
8. Any errors in the workflow are captured by Observability which sends an email to BaRS R&M team

4.2.2.2 Bulk ingestion and processing flow – DUEC DoS tactical solution – Update Endpoint Workflow

1. DUEC DoS team publishes the latest Master Switch Log (MSL) file to SharePoint
2. BaRS R&M team retrieves the MSL file from SharePoint
3. BaRS R&M team extracts data fields from MSL and creates a CSV file (important fields are ODS ID, DOS Service ID, Supplier Name)
4. BaRS R&M team uploads the CSV file to S3 bucket

5. File upload triggers the automated validation process by the “file ingestion” Lambda which does a validation to ensure presence of correct file format and mandatory fields
6. Next step involves the “data enrichment” lambda performing data enrichment per row in the CSV
7. The enriched data is sent to the API backend which performs the required operation (Insert/Update/Delete) on the endpoint catalogue in DynamoDB
8. Any errors in the workflow are captured by Observability which sends an email to BaRS R&M team

4.2.2.3 Delete Endpoint workflow

1. Supplier contacts NHSE BaRS team and specifies the endpoint(s) which need to be deleted from the EPC
2. NHSE BaRS team provides details to BaRS R&M team
3. BaRS R&M team creates a CSV and uploads it to S3 bucket
4. File upload triggers the automated validation process by the “file ingestion” Lambda which does a validation to ensure presence of correct file format and mandatory fields
5. The next step involves the EPC updating lambda which processes every row of the CSV file and sends details to the API backend
6. The API backend deletes the endpoint(s) / template from the catalogue in DynamoDB
7. Any errors in the workflow are captured by Observability which sends an email to BaRS R&M team

4.3 Deliverables

4.3.1 Build new Endpoint Catalogue (EPC)

- We will be building a new Endpoint Catalogue that will support use cases for real-time API requests from suppliers and receivers, via APIM, and various endpoint management scenarios as described in section 4.2.2.
- To begin, we will be creating a Proof of Concept of a standalone endpoint catalogue (Milestone M03). This will be deployed into the development environment, with basic functionality (API get request) and will include a minimum viable set of endpoints with mock/sample data.
- Following creation of the Proof of Concept, we will gather feedback and move on to building the full EPC (Milestone M04).
 - Components that make up the full EPC include:
 - AWS API gateway will be provisioned to communicate with the BaRS proxy on APIM
 - AWS Lambdas will be provisioned to form the EPC API backend
 - AWS DynamoDB will be provisioned as data store for the EPC
 - The EPC will undergo comprehensive testing (integration, system) and meet all functional and non-functional requirements.

- This will be deployed to dev, INT and prod environments.
- A tactical solution for onboarding and maintaining (DUEC DoS process) endpoints will be required and is mentioned further in section 4.3.2.
- The full endpoint catalogue will include delivery of a series of post endpoints for HealthcareServiceID and EndpointID. These are not currently used by the legacy EPC in production, but are required to support future self service requirements.

4.3.2 Tactical solution for bulk ingestion

- Within the scope of this stage of delivery for the BaRS EPC, there will not be an accompanying application with a User Interface for internal administration of the Endpoint information.
- Instead, we will build a tactical solution for ingestion and this will support scenarios for onboarding (as described in section 4.2.2) as well as the DUEC DoS Pharmacy process.
- This will be delivered alongside the full endpoint catalogue mentioned above (Milestone M04)
- Components that make up this solution include:
 - AWS S3 bucket will be provisioned for BaRS Run & Maintain team to upload the CSV file
 - AWS Lambdas will be provisioned for file validation and enrichment
 - AWS SQS will be used for message handling between file validation and enrichment Lambdas and APi backend for updating the data store in DynamoDB
- The following scenarios will be supported by this solution:
 - Onboarding and management processes:
 - Add endpoint template
 - Update endpoint template
 - Delete endpoint template
 - Create endpoint based on a template
 - Create endpoint without a template
 - Update existing endpoint
 - Delete an endpoint
 - DUEC DoS Pharmacy processes
 - Switch to a new endpoint

4.3.3 Updates to the BaRS Proxy

- To support our development of the BaRS EPC, we require some changes to be made to the BaRS Proxy that is hosted on APIGEE and supported by the APIM team.
- At a high level, the current version of the BaRS Proxy is using a JSON file from an S3 bucket. To support our solution, we will need to use an API with MTLS authentication and send headers for OrganisationID, ServiceID and EndpointID.
- Development changes to the BaRS Proxy on APIM will likely be made by the delivery team with support from APIM for SME impact assessment and peer review of the changes. This will be confirmed as part of lower level discussions with APIM team.

4.3.4 Migrate data from legacy EPC to new EPC

- Once we've developed the full EPC, we will need to migrate the data from current EPC (BaRS owned) to the new one.
- This data migration will need to be validated with no data loss or integrity issues post-migration.

4.4 Resilience & Recovery

- The EPC has a service level classification of Gold with the following resilience and recovery requirements: Appendix B - Service Classifications - NHS England Digital⁴
- Our primary requirements for Resilience & Recovery are:
 - The EPC is being built to provide 99.9% availability
 - The EPC will be supported by NHSE Service Management team and Accenture's Run and Maintain Team which operates 24x7x365
 - DR recovery target of 4 hours
- Disaster Recovery - The EPC components are being built with IAC in Terraform for quick recovery. The code and EPC contents in DynamoDB will be immutably backed up in an AWS Backup account.
- The EPC service will be added to the BCDR (Business Continuity and Disaster Recovery) document for existing ERS and Wayfinder services.

4.5 Observability

- We will develop the solution with observability in line with other Gold Service systems (e.g. A&G, eRS).
- The EPC will audit all information access attempts and actions, retaining all pertinent data therein.
- Logs will need to include:
 - Full request/responses (headers and payloads) - Persisted for 180 days.
 - Audit needs to retain who made a request, when and what the outcome was - Persisted indefinitely.
- Observability will be supported by the following components:
 - CloudTrail – Tracks AWS API activity for the resources, including S3 buckets, IAM roles/policies, Lambda functions, API Gateway, DynamoDB, SQS, VPC/networking, KMS, and CloudTrail/CloudWatch configuration itself.
 - CloudWatch – Collects Lambda logs and alarms, API Gateway access logs, CloudTrail logs, and VPC flow logs configured here.
 - AppDynamics/Dynatrace – Provides application performance monitoring and end-to-end tracing.

4. <https://digital.nhs.uk/services/cloud-centre-of-excellence/cloud-security-good-practice-guide/9.-appendix-b-service-classifications>

4.6 Solution design benefits

- **Operational simplicity** - The use of managed, serverless services removes the need to maintain servers or container platforms. This reduces operational overhead and simplifies deployment, monitoring, and support processes.
- **Elastic scalability** - The solution automatically scales in response to demand across both API traffic and bulk ingestion workloads, ensuring consistent performance during peak usage or large data processing events.
- **Resilience and reliability** - The separation of synchronous and asynchronous processing ensures system stability. Messaging queues provide workload buffering and retry capabilities, while durable storage ensures data can be reprocessed if needed.
- **Security and controlled external access** - All external access is managed through the API layer, ensuring consistent application of security controls, traffic management, and monitoring policies.
- **Cost efficiency** - The platform follows a consumption-based model where compute and processing costs scale with usage, making it cost-effective for workloads with variable demand patterns.
- **Extensibility and future evolution** - The modular and event-driven nature of the architecture allows new data sources, validation rules, or processing workflows to be added without significant redesign, supporting long-term solution evolution.

4.7 Out of scope

- Building a new BaRS proxy
 - Building a user interface for the onboarding or management of endpoints
-

5 Technology & Tooling

Purpose

This section outlines the intended technical scope for the delivery and operation of the service, including tooling, environments, infrastructure, and development standards. It provides a shared understanding of the technologies and approaches that are expected to be used across the delivery lifecycles.

The technologies described within this document have been selected to support secure, scalable, and maintainable service delivery aligned with NHS digital standards, security frameworks, and operational governance expectations. In addition to identifying the selected technologies, this document outlines the benefits they provide and explains why they have been chosen in the context of regulated healthcare service delivery.

The overall intent is to ensure that the service is delivered using platforms and tooling that enable strong security controls, repeatable and auditable deployments, operational transparency, and long-term sustainability.

Platform Access

- AWS and GitHub are expected to be used as the primary platforms for infrastructure hosting and source code management.
- Additional access may be required depending on the AWS accounts and environments in scope, including ERS, A&G, and BaRS. Access will be provisioned through established onboarding processes and will follow NHS Role-Based Access Control (RBAC) principles, ensuring least-privilege access aligned to delivery responsibilities.

Benefits

- The use of AWS and GitHub provides mature, enterprise-grade platforms with strong native security capabilities, integrated audit logging, and extensive automation support. Their widespread adoption reduces onboarding complexity and supports consistent engineering practices across teams.

Rationale for Selection

- AWS and GitHub were selected due to their strong security posture, operational maturity, and alignment with modern cloud-native delivery practices. Their adoption reduces delivery risk and supports long-term maintainability.

Logging, Monitoring and Observability

Observability Tooling

- Amazon CloudWatch is expected to be used as the primary monitoring, logging, and alerting platform.

Benefits

- Native AWS Integration
- CloudWatch integrates directly with AWS services without requiring additional agents or third-party collectors in most scenarios. This simplifies deployment, reduces configuration overhead, and ensures monitoring coverage is automatically applied to new AWS resources as they are provisioned. Native service telemetry also improves data reliability and reduces gaps in observability.
- Centralised Operational Visibility
- CloudWatch provides a unified platform for metrics, logs, and alerting, allowing teams to correlate infrastructure behaviour with application performance in a single location. This centralisation improves incident detection speed, reduces investigation time, and enables more efficient cross-team operational collaboration.

Rationale for Selection

- Alignment with AWS-First Architecture Strategy
- Because workloads are hosted within AWS, CloudWatch provides seamless monitoring coverage across serverless, container, and infrastructure-based services without requiring additional integration layers. This reduces operational complexity and supports consistent observability standards across environments.
- Focus on Actionable Alerting Signals
- Alerting will prioritise meaningful service health indicators such as sustained error rate increases, service degradation, and infrastructure failures rather than transient or self-healing events. This approach reduces alert fatigue, improves signal-to-noise ratio, and ensures operational teams can respond quickly to genuine service risks.

Deployment Automation

- Lambda deployment will be managed through a build pipeline that packages the Lambda artifacts and publishes them to an Amazon S3 bucket. The Terraform repository and deployment pipeline will reference these artifacts from S3 when provisioning or updating the Lambda resources. This approach provides an automated mechanism for delivering application artifacts to the infrastructure layer without relying on a GitOps repository.

Deployment Process

- The build pipeline is responsible for building, packaging, and versioning the Lambda artifacts before pushing them to the S3 bucket. The Terraform pipeline then retrieves the required artifact version from S3 and deploys or updates the Lambda functions as part of the infrastructure deployment process.

Benefits

- This approach provides clear separation between application build and infrastructure deployment responsibilities while maintaining an automated workflow. Storing versioned artifacts in S3 also provides traceability and supports repeatable deployments.

Rationale for Selection

- Using a build pipeline with S3-backed artifacts simplifies the deployment model while still supporting automation and version control of Lambda packages. It also aligns well with Terraform-based infrastructure management and avoids the operational overhead of maintaining a GitOps repository.

Secrets Management

- AWS Secrets Manager will be used as the primary secrets management solution for application and infrastructure credentials. Secrets will be centrally managed and accessed at runtime where possible to avoid storing sensitive values in source code or static configuration.

Benefits

- This approach supports automated secret rotation, centralised secret lifecycle management, and reduces the risk of credential exposure. Runtime secret injection reduces the risk of secrets being stored in configuration repositories or container images.

Rationale for Selection

- This model aligns with NHS security policy requirements and supports secure-by-design service delivery, reducing the operational and security risk associated with credential management.

Infrastructure as Code

- Terraform will be used for infrastructure provisioning and lifecycle management across AWS environments. All infrastructure definitions will be maintained in version-controlled repositories and deployed using controlled CI/CD workflows.
- Terraform state will be stored using an Amazon S3 backend with access controlled via role-based permissions. Where appropriate, state locking mechanisms will be implemented to prevent concurrent changes.

Benefits

- Infrastructure as Code supports repeatable and consistent environment provisioning, reduces manual configuration risk, and provides a full audit trail of infrastructure changes. It also supports rapid environment rebuild and disaster recovery scenarios.

Rationale for Selection

- Terraform is widely adopted across regulated cloud environments and provides a mature, well-supported framework for infrastructure standardisation and long-term maintainability.

CI/CD Pipelines

- GitHub Actions will be used as the primary CI/CD platform supporting automated build, test, security scanning, and deployment workflows aligned to organisational standards.

- Pipelines will be designed to support consistent build and deployment processes across environments and will enforce security and quality gates prior to deployment to higher environments.
- Security and quality tooling integrated into pipelines will include SonarQube, automated secret scanning, Amazon ECR vulnerability scanning, Dependabot, and Tenable scanning.

Benefits

- This approach enables automated validation of code quality and security posture early in the delivery lifecycle. It reduces manual review overhead and supports consistent enforcement of organisational development standards.

Rationale for Selection

- Embedding security and quality controls directly into CI/CD pipelines supports shift-left security practices and improves overall delivery reliability and compliance.

AWS Infrastructure Resilience

- Database backup and recovery will primarily use RDS snapshots, supported by AWS Backup services where appropriate. Backup policies will be aligned to recovery objectives and data protection requirements.

Benefits

- AWS-native backup tooling provides automated backup scheduling, point-in-time recovery capability, and secure storage. This reduces operational overhead and supports compliance and audit requirements.

Rationale for Selection

- Native AWS backup capabilities integrate directly with platform security, monitoring, and access controls, improving reliability while simplifying operational management.

Backend Development Standards

- TypeScript are the preferred backend languages for service development. TypeScript has strong ecosystems, mature tooling, and broad industry adoption.
- Services are expected to follow secure coding practices, consistent API design standards, and structured logging approaches to support maintainability and operational visibility.

Benefits

- TypeScript support rapid development, strong community support, and access to a large talent pool. Mature tooling supports automated testing, security scanning, and code quality validation.

Rationale for Selection

- TypeScript provides a balance between delivery speed, maintainability, and long-term supportability across teams.

Interoperability & FHIR

- FHIR implementation guidance will be followed where applicable to support integration with NHS and partner healthcare systems.

Benefits

- FHIR supports structured and standardised healthcare data exchange, improving interoperability and reducing integration complexity.

Rationale for Selection

- Alignment with NHS interoperability standards is essential for integration with national healthcare services and supports future system interoperability.

Technology Stack Summary

Layer / Capability	Technology / Tool	Usage / Notes
Cloud Platform	AWS	Primary cloud provider
Source Control	GitHub	Source code management
CI/CD	GitHub Actions	Build and deployment pipelines
Infrastructure as Code	Terraform	AWS infrastructure provisioning
Terraform State	Amazon S3	Centralised state backend
Secrets Management	AWS Secrets Manager	Centralised secrets store
Monitoring & Logging	Amazon CloudWatch	Metrics and logs
Static Code Analysis	SonarQube	Code quality and security
Build Artifact Storage	Amazon S3	Artifact Storage
Backend Language	TypeScript	Preferred implementation
Load Testing	JMeter / k6	Performance testing
Database	DynamoDB	Managed relational database

6 Quality & Assurance

The scope of our testing can be found in this sub page of the Scope Document: [BaRs EPCs API Test Strategy](#) (see page 31)

7 Deployment & Handover to Live Service

Deployment

- Deployment to production will follow the standard RAA processes and ways of working. documented here: [Release Process](#)⁵

Handover to Live Service

- The delivery team will provide a Technical Run Book for the new EPC and conduct knowledge transfer sessions with the BaRS Run & Maintain team, and any other teams who will be supporting the management of the live service.
- Knowledge transfer will cover the following topics
 - EPC architecture
 - EPC update mechanisms
 - Integration points with BaRS
 - Data migration process
 - Operational support procedures
- The below table documents the stakeholders that will be involved in live service support for the BaRS EPC.

Team name	Post Go-live role
NHSE Service Management Team	Level 1 support
Accenture Run & Maintain Team	• Level 2 & 3 support • Support DUEC DOS interim solution for EPC refresh • Support supplier onboarding
Accenture BaRS team	Level 3 support for EPC
NHSE APIM team	Level 3 support for BaRS proxy
DUEC DOS team	Updating Master Switch Log stored on SharePoint for EPC refresh

5. <https://nhsd-confluence.digital.nhs.uk/spaces/ERS/pages/1051872591/Release+Process>

8 Delivery Milestones

Mil est one Ref.	Milestone Description	Acceptance Criteria	Buyer Dependencies
M01	Reporting	• Bi-weekly reporting - Providing a reporting framework and to provide key MI/KPIs.	
M02	Scoping and backlog development	• A comprehensive scope document is agreed and signed off by key stakeholders. • A product backlog is created in the chosen Agile tool (e.g., Jira), containing user stories with clear descriptions, acceptance criteria, and prioritisation. • Dependencies and risks are identified and documented. • Initial sprint planning is completed with estimates for future sprints.	• Functional and non-functional requirements documented. • Dependency on coding language and technical tooling by Buyer
M03	Delivery of a proof-of-concept endpoint catalogue in dev environment	• A working prototype of the Endpoint Catalogue is deployed in the development environment. • The catalogue includes a minimum viable set of endpoints with mock or sample data. • Basic functionality (e.g., search, view, filter) is demonstrated. • Stakeholder feedback is collected and documented for iteration.	• Proof of concept architecture

Milestone Ref.	Milestone Description	Acceptance Criteria	Buyer Dependencies
M04	Fully tested and compliant Endpoint Catalogue	• All functional and non-functional requirements are met. • Catalogue passes integration, system testing and any other required testing. • Compliance checks (e.g., data protection, security standards) are completed and signed off. • Test results are documented and defects are resolved or triaged. • A tactical solution for DUEC DoS or others to maintain their endpoints e.g. CSV files emailed to an SES address that are picked up by a Lambda that does a couple of lookups and checks before pushing the data through the End Point API. To be able to add, remove and change endpoints.	• Availability of resource to make changes of UEC DoS to onboard to the API • Technology and framework selection. Dependency on buyer for coding languages and technical tooling. • There is a dependency that if the UEC DoS are not able to update the endpoint catalogue then there is a need for a user interface or other tactical solution to enable them to update endpoints.
M05	Development changes to the BaRS Proxy: • The BaRS Proxy needs to utilise the new EPC which will require development changes to the BaRS Proxy and a plan to deploy this code change to Production, in coordination with the BaRS Team • Migration of data from current EPC (BaRS owned) to the new one developed.	• BaRS Proxy is updated to interact with the new Endpoint Catalogue. • Code changes are reviewed, tested, and approved. • A deployment plan is created and agreed with the BaRS, APIM and Live Service teams. • Data migration from the legacy EPC to the new EPC is completed and validated. • No data loss or integrity issues post-migration.	• Timely accessibility to technical resource from the APIM team currently managing the Proxy.
M06	Delivery of the full Endpoint Catalogue in live environment	• Endpoint Catalogue is deployed to the live environment. • Live deployment is verified with smoke testing. • Monitoring and logging are enabled for live operations.	

Milestone Ref.	Milestone Description	Acceptance Criteria	Buyer Dependencies
M07	Test full EPC solution with INT BaRS Proxy	- Rollback plan is in place and tested - Integration testing between EPC and INT BaRS Proxy is completed. - All test cases pass, including edge cases and failure scenarios. - Performance benchmarks are met. - Test results are documented and signed off by the BaRS team.	Updated BaRS Proxy from M05
M08	Handover complete to Live Services	Technical Run Book completed and knowledge transfer session to service analyst.	Accessibility of live service staff for knowledge transfer

9 Stakeholders

NHSE

- Pete Ward - Deputy Director of Delivery
- Debbie McAuliffe - Head of Product
- Candice Moore - Head of Delivery
- Craig Sawyer - Principal Systems Engineer
- Alek Radjenovic - Senior Solution Architect
- Jamil Aziz - Senior Delivery Manager
- Leon Finnerty - Lead Product Manager
- Richard Ward - Solution Architect
- Carl De'ath - Technical Business Analyst
- Stephen Potter - Senior Service Manager

Accenture

- Guy Player - Managing Director
 - Sorcha Cotter - Senior Technical Architect
 - Hozefa Jamali - Delivery Manager
 - Indar Jagdev - Technical Business Analyst
 - Mukhtar Ata - Lead DevOps Engineer
 - Shanthi Thakku Venkatesh - Senior Quality Engineer
 - Alekhya Sunkavalli - Senior API Engineer
-

10 Risks, Assumptions, Issues & Dependencies

The full RAID log is stored on SharePoint here: [RAID Log - BaRS Endpoint Catalogue Delivery v1.0.xlsx](#)⁶

The full raid log is updated on a regular basis. The RAID items below are a view at time of sign off of this document.

Summary of open RAID items:

ID	Category	Description	Mitigation / Comments	Status	RAG	Raiser
R.01	Risk	There is a risk that if the delivery overruns the expected end date of the 31st March 2026 into the next financial year there will be an impact due to budgeting constraints.	- It will not be possible to complete all delivery milestones by the end of March. - Agreed with Jamil Aziz and Nicola Catto to raise a change note to reflect updated delivery timelines. - Recovery actions are being implemented to bring delivery dates in. - Producing a delivery plan to manage dependencies with other teams (DUEC DoS, APIM, BaRS, Service Management).	Open	AMBER	Originated from signed SoW

6. https://nhs.sharepoint.com/:x:/r/sites/msteams_d0a2ea-BaRSEndpointCatalogue/Shared%20Documents/BaRS%20Endpoint%20Catalogue/Governance/RAID%20Log%20-%20BaRS%20Endpoint%20Catalogue%20Delivery%20v1.0.xlsx?d=w382b71c925514e6c8d1aa34581aff82b&csf=1&web=1&e=saf4U9

ID	Category	Description	Mitigation / Comments	Status	RAG	Raiser
R.02	Risk	The DUEC DoS team do not have agreed development or operational resource available from 1st April and may not have capacity to support before then. This introduces a risk that they will not be able to support with the development of a tactical solution that will allow them to update the new EPC. The possible impacts of this are: 1) additional work for our delivery team, 2) continued involvement and work from the BaRS run & maintain team, 3) delays to the delivery timelines	- Delivery team are scoping out options for the tactical solution including what input is required from various teams (EPC Delivery Team, BaRS Run & Maintain, DUEC DoS) 10/02 – Team now has a proposed option which is being ratified with all parties before progressing. 17/02 - This option would not require any development changes or business process changes for the DUEC DoS team however it does require more design and build work for the delivery team. 02/03 - Agreed the proposal with BaRS team and now need to loop back with DUEC DoS to run them through and agree.	Open	AMBER	Owen Leyshon
D.01	Dependency	The Buyer will be responsible for providing the Supplier with access to its systems, Subject Matter Experts, and relevant documentation / business intelligence including that held by Third Parties. In the event of issues arising refer to Schedule 13A section 6.	Jamil has provided support with engaging with DUEC DoS, APIM and Service Management teams.	Open	GREEN	Originated from signed SoW
D.02	Dependency	The parties shall work together to develop mutually acceptable acceptance criteria and document an agreed timebound process. Unless and until agreed, the sole criteria for acceptance of Milestones shall be material conformance with the item description.	- Milestone process call attended on 30/01 - Need to review & agree with Jamil and other relevant stakeholders if any changes to the Milestones and Acceptance Criteria are required as part of our Scope Document sign off. - Need to agree the evidence for each Milestone.	Open	GREEN	Originated from signed SoW

ID	Cat egory	Description	Mitigation / Comments	St at us	RAG	Raiser
D .0 4	Dep ende ncy	Technology and framework selection. Dependency on buyer for coding languages and technical tooling.	- Delivery team engaged with existing Accenture teams in eRS & A&G to understand current technology and tooling. - Proposal has been documented as part of Scope Document and is being reviewed with Buyer.	Op en		Originat ed from signed SoW
D .0 6	Dep ende ncy	Availability of resource to make changes of UEC DoS to onboard to the API	04/02 - During out introduction call on 03/02, DUEC DoS team were not aware of our scope and we are now needing to work through a series of options, understand next steps and any dependencies. They do not currently have any development or operational resource available to support this. 17/02 – Team now has a proposed option which is being ratified with all parties before progressing. This option would not require any development changes or business process changes for the DUEC DoS team. 02/03 - Agreed the proposal with BaRS team and now need to loop back with DUEC DoS to run them through and agree.	Op en		Originat ed from signed SoW
D .0 7	Dep ende ncy	There is a dependency that if the UEC DoS are not able to update the endpoint catalogue then there is a need for a user interface or other tactical solution to enable them to update endpoints.	- This tactical solution is being documented as part of Scope Document and reviewed with Buyer. 17/02 – Team now has a proposed option which is being ratified with all parties before progressing. This option would not require any development changes or business process changes for the DUEC DoS team. This tactical solution is being documented as part of Scope Document and reviewed with Buyer 02/03 - Agreed the proposal with BaRS team and now need to loop back with DUEC DoS to run them through and agree.	Op en		Originat ed from signed SoW

ID	Category	Description	Mitigation / Comments	Status	RAG	Raiser
D.08	Dependency	Timely accessibility to technical resource from the APIM team currently managing the Proxy.	• Introduction call with APIM team planned for Thursday 12/02. Was originally planned for Monday 02/02 but needed to be rescheduled due to APIM availability. • 17/02 - Productive call held with the APIM Team. They were not aware of our scope however the change is straightforward and so they are able to support us within our project timeframe. • 02/03 - We now require changes to the BaRS Proxy to support our EPC solution so re-engaging with APIM to discuss the new requirements. • 11/03 - Have had a conversation with APIM and it's very likely that Accenture will need to perform the development changes with SME support from APIM. Lower level discussions and planning are now progressing.	Open	AMBER	Originated from signed SoW
D.09	Dependency	Accessibility of live service staff for knowledge transfer	• Jamil has shared details of key contacts so Owen can arrange introduction call. • 17/02 - Productive call held on 16/02 to introduce our project and understand next steps. Need to complete actions from that call including completing a new work request and following the process to onboard a new service.	Open	GREEN	Originated from signed SoW
D.10	Dependency	Dependency on completing a DPIA	• Jamil is going to provide us with instructions on how to complete this • 17/02 - This should now be possible to progress following engagement with the Service Management team.	Open	GREEN	Jamil Aziz

11 BaRs EPCs API Test Strategy

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11.1 1 Introduction

The purpose of this strategy is to ensure the reliability and performance of the BaRs Endpoint Catalogue Service (EPCSe). APIs serve as the backbone of our distributed systems, requiring a shift-left approach where testing begins at the design phase.

11.1.1 1.1 Project Overview

The Endpoint Catalogue Service (EPCSe) is required to facilitate the routing of standard-compliant messages (such as BaRS or ITK) within the NHS by providing a mechanism to look up and use correct digital endpoints for healthcare services. The EPCSe provides for a more flexible, reliable, and robust management of healthcare services and the systems that support them and moves away from manual, lengthy processes prone to error.

The service will operate using an API-first approach, emphasising integration with NHS systems via accessible APIs. It offers a self-service model, enabling users to manage endpoints, healthcare services, and their relationships in a decentralised fashion. The data model supports identifying and organising healthcare services and their digital endpoints through unique identifiers, ensuring accuracy and flexibility in routing system-to-system communication.

The EPCSe will mostly be used (either by the sending system or an intermediary service) as a look-up service to find the correct address (URL) of a standard-compliant endpoint where the message should be forwarded to.

11.1.2 1.2 Primary Objective

To validate that all endpoints return accurate, predictable results and maintain data integrity under peak loads.

11.2 2. Scope of Testing

11.2.1 2.1 In-Scope

- **Functional Logic:** Validation of all REST endpoints against technical specifications.
- **Contractual Integrity:** Verification of JSON response structures against defined schemas.
- **Security & Auth:** Validation of JWT authentication layer.
- **Performance Benchmarking:** Performance benchmarks are validated using JMeter with real-time and bulk data load testing.
- **Error Handling:** Validation of standard HTTP status codes for malformed requests and unauthorized access.
- **Data Persistence:** Verification of downstream database state changes following successful API transactions.

11.2.2 2.2 Out-of-Scope

- **User Interface (UI):** Validation of browser-based front-end components.
- **Internal Service Logic:** The internal code architecture of third-party external providers.
- **Infrastructure Stress:** Testing of underlying physical server hardware or network cabling.

11.3 3. Test Approach & Methodologies

11.3.1 3.1 Functional testing

API Functional Testing validates that the software's **business logic** behaves exactly as defined in the requirements. In this project, it focuses on verifying that the API accepts specific inputs and returns the correct outputs, status codes, and data structures, ensuring the backend processes requests accurately without a user interface.

11.3.2 3.2 Functional Test Types

We will execute the following test types to ensure full coverage:

- **Smoke Testing:** Verifying critical endpoints (e.g., Auth, Health Check) are "Up" after a deployment.
- **Positive (Happy Path) Testing:** Using valid data to ensure the API performs its primary function (e.g., successful payment).
- **Negative Testing:** Using invalid/malformed data to ensure the API returns correct 4xx error codes and doesn't crash.
- **Boundary Testing:** Testing limit values (e.g., a "quantity" of 0 or the maximum allowed characters in a name field).
- **Integration Testing:** Verifying the flow of data between this API, Bars Proxy and the database.
- **End-to-End (E2E) Testing:** Validating a complete user journey

11.3.3 3.3 Assertion Standards

This section defines the "Pass/Fail" logic for all API automated scripts. Use the following categories for a real-time project:

- **HTTP Status Code Assertions**
 - **Success:** 200 OK for GET/PUT, 201 Created for POST, 204 No Content for DELETE.
- **Client Errors:**
 - 400 Bad Request for invalid payloads, 401 Unauthorized for missing tokens, 403 Forbidden for insufficient permissions.
- **Response Header Assertions**
 - Validate security and metadata:
 - Verify Content-Type is application/fhir+json?.
 - Ensure security headers like Strict-Transport-Security or X-Content-Type-Options are present.
- **Schema Validation (The "Contract")**
 - Assert that the response body strictly matches the **Swagger/OpenAPI** specification.
 - Validate data types
 - Check for required fields and ensure no unexpected fields are returned.

- **Data Integrity & Business Logic**
- **Value Matching:**
 - Assert that the returned ID matches the one sent in the request.
- **State Verification:**
 - For a DELETE call, the assertion should include a follow-up GET call to verify the resource returns a 404 Not Found.

11.4 4. Performance Testing

This section defines how the API's responsiveness, stability, and scalability will be validated under traffic patterns.

Objectives: Identify bottlenecks, determine the maximum throughput (TPS), and ensure the API adheres to **Service Level Agreements (SLAs)**.

11.4.1 4.1 Performance Metrics (KPIs)

- **Response Time:** 95th Percentile (P95) should be < 300ms.
- **Throughput:** Minimum 500 Transactions Per Second (TPS).
- **Error Rate:** Less than 0.1% under peak load.
- **Resource Utilization:** CPU usage < 70%, Memory leaks < 1% over 24 hours.

11.4.2 4.2 Test Types

- **Load Testing:** Testing with the expected volume (e.g., 1000 concurrent users).
- **Stress Testing:** Pushing the API beyond its limits to identify the "breaking point."
- **Soak/Endurance Testing:** Running a steady load for 12+ hours to check for memory leaks.
- **Spike Testing:** Sudden jumps in traffic (e.g., "Flash Sale" scenario).

11.4.3 4.3 Tools

- **JMeter** or **k6**

11.5 5. Test Data Management

11.5.1 5.1 Objectives & Scope

- **Purpose:** Define the goal, which is usually ensuring timely access to high-quality, compliant test data.
- **Systems Covered:** List which databases (SQL, NoSQL) and APIs fall under this management policy.

11.5.2 5.2 Data Sourcing & Creation

- **Static/Dummy Data:** Data used for basic smoke tests that is hardcoded or foundational.
- **Synthetic Data:** Artificially generated data for new features or scenarios requiring large volumes of unique records.
- **Production Sub setting:** Extracting smaller, relevant data sets from live systems to preserve realism.
- **Mocked Responses:** To simulate data from external, third-party APIs. (WireMock)

11.5.3 5.3 Data Maintenance & Lifecycle

- **Provisioning Automation:** Describing the automated delivery of fresh data to testing environments.
- **Refresh Schedule:** A defined frequency (e.g., "daily automated database reset") for updating stale data.
- **Cleanup Procedures:** Procedures for removing temporary test data created during automated runs (e.g., using DELETE API calls or database scripts in the test teardown phase).
- **Data Archiving:** Policy for storing old test data securely.

11.6 6. Environment & Tooling

11.6.1 6.1 Environment Infrastructure

- **Dev:** Where developers perform initial unit and contract tests.
- **QA/Staging:** A dedicated environment that mirrors **Production** (same CPU/RAM) for functional and performance testing.
- **Performance Environment:** A scaled or isolated environment specifically for **Load/Stress** tests to avoid impacting other QA activities.
- **Base URLs:** Document the endpoints

11.6.2 6.2 Tools

- **Functional Automation: Rest Assured** (Playwright using JavaScript) and **Postman**.
- **Performance/Load: K6** or **JMeter**.
- **Security Testing: OWASP ZAP** for vulnerability scanning.
- **Contract Testing: Pact** or **Postman API Governance** to ensure schema consistency.

11.6.3 6.3 CI/CD Integration

- **Pipeline Tool:** GitHub Actions.

- **Triggers:** "Run smoke suite on every Pull Request; run full regression on every merge to the main branch."
- **Failure Policy:** Define if a failed API test should "Break the Build" (Stop deployment).

11.6.4 6.4 Reporting

How results are communicated to stakeholders:

- **Dashboards:** Allure Report or Cucumber Reports (From API test automation framework)

11.7 7. Defect Management & Metrics

11.7.1 7.1 Defect Tracking Tool

- All defects will be logged in **Jira** using the "Bug" issue type within the EPCSe-API project dashboard.

11.7.2 7.2 Defect Lifecycle Workflow

Outline the clear stages a bug goes through from creation to closure.

- **New:** The bug has been logged by the QA team.
- **Open/Assigned:** The development team lead has acknowledged the bug and assigned it to a developer.
- **In Progress:** The developer is actively working on a fix.
- **Fixed/Resolved:** The fix has been committed to a branch.
- **Verified (Retested):** QA has confirmed the fix works in the testing environment.
- **Closed:** The fix is in production and verified.
- **Rejected/Deferred:** The bug is not actually a bug or will be fixed in a future release.

11.7.3 7.3 Defect Reporting Requirements

Define mandatory fields that must be included when a QA engineer logs a bug to avoid back-and-forth communication.

- **Mandatory Fields:**
 - **Summary:** A clear, concise title.
 - **Steps to Reproduce:** Bulleted list of API calls/actions.
 - **Expected Result:** What the API *should* do.
 - **Actual Result:** What the API *did* (including exact JSON response/status code).
 - **Environment:** The URL where the bug was found.

- **Payload/Request Body:** The exact data used.
- **Attachments:** Screenshots of Postman console/logs.

11.8 8. Risk Management & Contingency Plan

11.8.1 8.1 Technical Risks

- **API Unavailability:** Downtime of internal microservices or third-party sandboxes during testing.
- **Data Integrity:** Instability of the test database causing false negatives.

11.8.2 8.2 Project Risks

- **Scope Creep:** Frequent changes to API contracts mid-sprint.
- **Environment Delays:** Delays in the availability of the staging or performance environment.
- **Test Data Availability:**

11.9 9. Approval & Sign-off

11.9.1 9.1 Document Control

Before the signature table, include a versioning block to track changes.

- **Version:** v1.0.0
- **Status:** (Draft / Under Review / Approved)
- **Last Updated:** Jan 26, 2026

11.9.2 9.2 The Approval Matrix

This section confirms that the API Test Strategy has been reviewed and approved by all relevant stakeholders. Approval indicates agreement on the testing approach, scope, tools, timelines, and quality criteria defined in this document.

Role	Name	Responsibility	Status
Delivery Manger	Leyshon, Owen D	Responsible for overseeing end-to-end service delivery, ensuring quality, compliance, and stakeholder satisfaction.	Approved/ Pending
Technical BA	Jagdev, Indar	Ensures successful delivery of the service on time, within scope, and meeting all standards.	Approved/ Pending

Devops	Ata, Mukhtar	Ensures reliable, secure, and compliant deployment of services.	Approved/ Pending
Developer	Sunkavalli, Alekhya	Responsible for building secure, reliable, and standards-compliant APIs.	Approved/ Pending
QA	Shanthi Thakku Venkatesh	Ensures all deliverables meet quality, safety, and compliance standards.	Approved/ Pending

11.9.3 9.3 Entry and Exit Criteria

Entry Criteria	Exit Criteria
Requirements and test strategy are finalized and approved	All planned test cases are executed
Application/APIs are deployed in the test environment	No open Critical or High severity defects
Test cases, test data, and tools/frameworks are ready	Test coverage and pass criteria are met
Test environment is stable and accessible	Performance/SLA requirements are achieved (if applicable)
Required access, credentials, and dependencies are available	Test reports are shared and stakeholder sign-off is received

12 Observability - High Level Breakdown

Observability high level summary

Observability Implementation Summary

Executive Overview

We implemented a comprehensive observability platform across two infrastructure repositories using CloudWatch, OpenTelemetry, and Terraform infrastructure-as-code. The system enables real-time detection of issues, rapid diagnosis through distributed tracing, and intelligent alert routing based on severity levels.

Part 1: Unified CloudWatch Platform

What Was Done

Established AWS CloudWatch as the single observability hub for all logs, metrics, and alarms across both the Catalog API and Data Ingestion platforms.

Best Practice Alignment

Single Source of Truth: Rather than fragmenting observability across multiple tools (one for logs, another for traces, another for metrics), centralizing on CloudWatch eliminates context switching during incident response.

Native AWS Integration: CloudWatch integrates natively with Lambda, API Gateway, DynamoDB, and SQS—the core services in use. This eliminates the operational overhead of managing external collection agents.

Cost Efficiency: Leveraging native AWS services avoids additional vendor licensing and reduces the number of systems the team must maintain.

Value Added

Engineers access logs, metrics, and alarms from a single interface

No custom log shipping agents required

Cost savings from avoiding third-party observability platforms

Faster incident response without cross-tool navigation

Part 2: Severity-Based Alert Routing

What Was Done

Created five priority-level SNS topics (p1 through p5) that automatically route alarms to appropriate notification channels based on criticality.

P1 (Critical): Infrastructure down, data loss risk → immediate notification to on-call

P2 (High): Service degradation → urgent but less critical

P3 (Medium): Functional issues with workarounds

P4 (Low): Non-critical errors

P5 (Info): Status updates and routine operations

Best Practice Alignment

Alert Fatigue Prevention: Organizations that send all alerts with equal urgency experience alert fatigue—engineers learn to ignore alerts, causing them to miss critical issues. Severity-based routing ensures engineers pay attention to signals that matter.

Appropriate Escalation: Critical infrastructure failures need immediate response; informational events can accumulate in background channels. Matching urgency to impact prevents both over-alerting and under-alerting.

Team Specialization: Operations teams subscribe to infrastructure alerts (p1-p3) while development teams monitor application errors (p3-p5). Each team receives relevant signals without noise.

Value Added

On-call engineers don't ignore alerts due to noise—each alert carries proportional urgency

Critical issues reach teams immediately through multiple channels (email, Teams)

Teams can customize subscriptions without infrastructure changes

Low-priority events don't interrupt but remain searchable for analysis

Part 3: Domain-Specific Dashboards

What Was Done

Created five specialized CloudWatch dashboards, each focusing on a specific operational domain:

API Gateway Dashboard: Request volume, latency percentiles, error rates, throttling

DynamoDB Dashboard: Capacity consumption, throttling events, query patterns

Lambda Performance Dashboard: Execution duration, cold starts, concurrency

Health and Business Metrics Dashboard: Application-level KPIs and health indicators

Aggregate Performance Dashboard: System-wide health status at a glance

Best Practice Alignment

Separation of Concerns: Rather than a single dashboard mixing infrastructure, application, and business metrics, domain-specific dashboards allow operators to focus on their area of responsibility. This reduces cognitive load and enables faster diagnosis.

Answerable Questions: Each dashboard answers a specific operational question without requiring cross-domain expertise:

"Is the API performing?" → API Gateway Dashboard

"Are we hitting database limits?" → DynamoDB Dashboard

"Are Lambda functions efficient?" → Lambda Performance Dashboard

Role-Based Views: API teams focus on their dashboard, database operators on theirs. This enables specialization without requiring everyone to understand every system.

Value Added

Operators answer operational questions in seconds without log analysis

Different teams (API, database, platform) can own their observability

New team members learn system behavior from visual dashboards

Pre-built dashboards eliminate the "what metrics should I look at?" decision

Part 4: OpenTelemetry Instrumentation

What Was Done

Integrated AWS Distro for OpenTelemetry (ADOT) into Lambda functions through:

ADOT Lambda layer references for automatic instrumentation

Environment variables enabling distributed tracing (AWS_LAMBDA_EXEC_WRAPPER)

Service names per function for trace identification

Applied to both API Lambda functions and Data Ingestion pipeline functions

Best Practice Alignment

Vendor Neutrality: OpenTelemetry is becoming the industry standard for observability. Using ADOT aligns with this standard rather than locking into proprietary solutions.

Zero-Code Instrumentation: Rather than embedding custom tracing libraries in application code, ADOT provides automatic instrumentation through Lambda wrapper configuration. This means:

No code changes to application

Instrumentation updates happen through layer updates

All functions automatically gain tracing capabilities

Standards-Based Approach: OpenTelemetry is vendor-agnostic, enabling future migration to other observability backends without rewriting instrumentation.

Value Added

Distributed tracing shows request execution path across services and timing at each step

Engineers identify bottlenecks (is slowness in application logic or external API?)

Root cause analysis is visible through traces—no log parsing required

Future observability platform changes don't require application code rewrites

Part 5: Application-Level Error Detection

What Was Done

Configured CloudWatch Logs metric filters that parse application logs and detect specific error patterns:

Monitors for structured log entries indicating failures

Detects "Error" messages paired with "EPC API" references

Converts matching log entries to CloudWatch metrics

Metrics trigger alarms when error threshold exceeded

Best Practice Alignment

Business-Level Visibility: Infrastructure metrics (CPU, memory, latency) show that something is wrong.

Application metrics show what is wrong and which business process is affected.

Structured Logging Pattern: Using structured (JSON) logs enables pattern matching and field extraction. This is superior to free-form text logs where parsing is fragile.

Early Detection: Metric filters process logs in real-time, enabling alarms to trigger on the first error rather than discovering issues through log analysis hours later.

Value Added

Distinguishes between internal failures and upstream API failures

Detects application errors that infrastructure metrics would miss

Enables rapid escalation to upstream teams when external dependencies fail

Provides business context ("ingestion processing failed") rather than just infrastructure context

Part 6: Email and Teams Alert Delivery

What Was Done

Connected CloudWatch alarms to notification channels:

Email subscriptions to SNS topics route alerts to distribution lists

Lambda function subscribes to SNS topics and posts formatted messages to Teams channels

Different severity levels route to different channels

Best Practice Alignment

Reliable Notification: Email ensures notification arrives even if network connectivity is degraded.

Teams provides real-time team awareness. Using both ensures no single point of failure in alerting.

Searchable Audit Trail: Both email and Teams create searchable records of all alerts, supporting post-incident analysis and compliance requirements.

Non-Intrusive Yet Visible: Lower-priority alerts accumulate in email for batch review. Critical alerts trigger Teams notifications that interrupt the team, preventing both over-alerting and under-alerting.

Value Added

On-call engineers receive alerts through reliable channels they actively monitor

Alert history is searchable for root cause analysis

Team awareness without requiring everyone to monitor infrastructure

Audit trail of system alerts for compliance and historical analysis

Part 7: Infrastructure as Code Discipline

What Was Done

Defined all observability infrastructure (dashboards, alarms, Lambda functions, SNS topics) as Terraform modules with:

Explicit `required_version` and `required_providers` blocks

Modular structure where each domain has independent modules

Version control for all observability changes

Infrastructure deployable to dev, int, and production environments

Best Practice Alignment

Reproducibility: Code-defined infrastructure deploys identically across environments. Manual AWS

Console configuration creates drift—different dashboards in dev vs. production, inconsistent alarm thresholds.

Change Tracking: Git history shows who changed alerting thresholds and when. This supports troubleshooting if configuration changes introduce issues and enables peer review before deployment.

Consistency: The same Terraform module instantiates in dev, int, and production, ensuring visual and behavioral consistency across environments.

Versioning: Terraform versioning requirements prevent "works on my version" problems where different provider versions produce different behavior.

Value Added

Rollback capability if alerting changes cause problems

Pull request workflow enables peer review of observability changes

Team members propose changes without direct AWS access

Production observability matches dev configuration exactly

Part 8: Development Environment Observability

What Was Done

Implemented complete observability stack in the Catalog API development environment:

Five domain-specific dashboards

Severity-based SNS alert routing

Email and Teams notifications

Custom metric filters for application errors

Why This Approach

Developer Feedback Loop: Developers see operational visibility while building. Rather than discovering observability gaps in production, gaps surface during development.

Consistency Testing: Verifying dashboards, alert logic, and notification routing work correctly in dev before production deployment reduces production issues.

Team Familiarity: Development team becomes fluent with observability tooling in a low-stakes environment, reducing operational mistakes in production.

Value Added

Development team owns their observability during feature development

Issues discovered and fixed before reaching production

Team learns operational context alongside application development

Part 9: Integration Environment Validation

What Was Done

Deployed the complete observability stack to the integration environment with production-like configurations:

Same dashboard layouts as will be used in production

Alert thresholds matching production expectations

Production-realistic metric retention and log retention periods

Stress-tested alert routing and notification delivery

Why This Approach

Production Dress Rehearsal: Integration environment mirrors production architecture. Testing observability in int validates that alerting strategies work correctly under production-like load before actual production deployment.

Alert Logic Validation: Testing alerts in int reveals if thresholds are too aggressive (over-alerting) or too lenient (under-alerting) before production is affected.

Team Readiness: Operations team practices incident response using production observability in int, building muscle memory before handling production incidents.

Value Added

Production observability validated before deployment

Alert thresholds proven appropriate under realistic load

Operations team trained on production alerting without affecting live systems

Configuration proven consistent between dev and int

Part 10: Distributed Tracing Across Data Pipeline

What Was Done

Instrumented the data ingestion pipeline with OpenTelemetry:

CSV ingestion Lambda captures file upload, validation, error handling

SQS processor Lambda captures queue consumption, message processing duration

Each service has distinct service name for trace identification

Traces show execution path from file upload through processing completion

Why This Approach

End-to-End Visibility: Rather than debugging each service independently, traces show the complete journey. When data processing fails, traces immediately show which pipeline stage failed and how long each stage took.

Bottleneck Identification: During high-volume ingestion, traces show whether slowness occurs in upload, validation, queue waiting, or processing. This distinction guides optimization efforts.

Upstream Dependency Monitoring: Traces capture interactions with external EPC API, showing whether delays or failures originate internally or from upstream dependencies.

Value Added

Data processing issues visible through traces without log analysis

Performance optimization targets identified through execution timing

Upstream API failures distinguished from internal processing failures

Complete audit trail of data through processing pipeline

Part 11: Strategic Outcomes

Operational Maturity

What Changed: Platform transitions from reactive (discovering issues through customer impact) to proactive (detecting issues automatically and diagnosing automatically).

How It Matters:

Issues detected in seconds rather than hours

Root causes identified through traces and dashboards rather than investigation

Preventive scaling based on trends rather than emergency scaling after hitting limits

Team Effectiveness

What Changed: Observability provides clear visibility, reducing expertise silos.

How It Matters:

New team members learn system behavior from dashboards without requiring mentoring

On-call rotations distribute evenly because observability reduces reliance on specific experts

Data-driven conversations replace opinion-based debates about performance

Cost Optimization

What Changed: Capacity decisions data-driven rather than guess-based.

How It Matters:

Right-sizing prevents over-provisioning and unnecessary costs

Trend analysis guides scaling during maintenance windows rather than emergency scaling

Alert severity routing prevents alert fatigue that leads to ignored notifications

Risk Mitigation

What Changed: Multiple detection mechanisms prevent silent failures.

How It Matters:

Custom metric filters detect application failures that infrastructure metrics would miss

Severity-based routing prevents critical issues being overlooked

Audit trails support compliance requirements

Early detection enables containment before cascading impact

Conclusion

The observability implementation establishes a strategic foundation for operational excellence. Rather than adding observability reactively when problems arise, we designed integrated systems answering key operational questions upfront:

Is the system healthy? (Aggregate Dashboard)

Where are bottlenecks? (Domain-specific Dashboards)

What caused this failure? (Distributed Traces)

Is this issue critical? (Severity-based Routing)

The modular architecture enables teams to own their observability. The OpenTelemetry instrumentation provides industry-standard distributed tracing. The severity-based routing ensures appropriate teams receive relevant alerts. The infrastructure-as-code approach enables consistent, reproducible observability across environments.

As the platform grows in complexity, this observability foundation scales with it, supporting increasingly sophisticated operational scenarios while maintaining simplicity through domain-specific views and standardized tools.

High level

Account ID

Bucket names are added as secrets

Variables

Secrets Manager

AWS secrets manager variable secret

Copying lambda manifest

Secret names

Variables

AWS secret management

13 Requirements List

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .1	A Self service methodology	FR	 RAA-3982⁷ - Requirement EPC.1 - A Self service methodology IDENTIFIED		API -> Business Logic / service controller -> Database/ repository	
EPC .2	API First Approach	FR	 RAA-4004⁸ - Requirement EPC.2 - API First Approach IDENTIFIED			

7. <https://nhsd-jira.digital.nhs.uk/browse/RAA-3982?src=confmacro>

8. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4004?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A ls o S a t i s f i e s	Notes	Comments / Questions
EPC .2.1	Appropriate Technology must be used to facilitate the interaction with Endpoint information.	FR	☒ RAA-4085⁹ - Requirement EPC.2.1 - Appropriate Technology must be used to facilitate the interaction with Endpoint information IDENTIFIED			
EPC .3	Internet-facing	FR	☒ RAA-4026¹⁰ - Requirement EPC.3 - Internet facing IDENTIFIED			

9. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4085?src=confmacro>

10. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4026?src=confmacro>

Requirement ID	Description	Type (FR / NFR)	Jira Link	Also Satisfies	Notes	Comments / Questions
EPC.4	Conforms to technical standards	FR	 RAA-4027 ¹¹ - Requirement EPC.4 - Conforms to technical standards IDENTIFIED		- Synchronous - Restful - Valid FHIR R4 - Uses appropriate security patterns.	
EPC.4.1	Conforms to technical standards - As a BaRS implementor I must use FHIR R4 resources to adhere to the standard.	FR	 RAA-4028 ¹² - Requirement EPC.4.1 - Conforms to technical standards IDENTIFIED			

11. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4027?src=confmacro>

12. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4028?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .5	Endpoint information Must be accessible, using business identifiers to search for and access information.	FR	 RAA-4030 ¹³ - Requirement EPC.5 - Endpoint information Must be accessible, using business identifiers to search for and access information. IDENTIFIED			

13. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4030?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A ls o S a t i s f i e s	Notes	Comments / Questions
EPC .5.1	Endpoint Information must be agnostic of Service Discovery	FR	 RAA-4031 ¹⁴ - Requirement EPC.5.1 - Endpoint Information must be agnostic of Service Discovery IDENTIFIED	E P C. 7	given dos id i and "BaRS", respond with corresponding endpoint information	

14. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4031?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .5.2	On request the solution must provide the current address for an Endpoint identified by the provided business identifiers	FR	 RAA-4032 ¹⁵ - Requirement EPC.5.2 - On request the solution must provide the current address for an Endpoint identified by the provided business identifiers IDENTIFIED			

15. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4032?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .5.3	An Endpoint is uniquely identified by the combination of multiple business identifiers	FR	 RAA-4033 ¹⁶ - Requirement EPC.5.3 - An Endpoint is uniquely identified by the combination of multiple business identifiers IDENTIFIED			

16. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4033?src=confmacro>

Requirement ID	Description	Type (FR / NFR)	Jira Link	Also Satisfies	Notes	Comments / Questions
EPC.5.3.1	The solution must be able to define a destination for requests by a service. The clinical service being requested - Clinical Service Identifier, agnostic of type	FR	☒ RAA-4035¹⁷ - Requirement EPC.5.3.1 - The solution must be able to define a destination for requests by a service. IDENTIFIED			
EPC.5.3.2	The solution must be able to define the method used to discover a service. The service discovery method used - Service Discovery Identifier, agnostic of type	FR	☒ RAA-4036¹⁸ - Requirement EPC.5.3.2 - The solution must be able to define the method used to discover a service. IDENTIFIED			

17. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4035?src=confmacro>

18. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4036?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .5.3 .3	The routing service being used - Routing Service Identifier	FR	 RAA-4037 ¹⁹ - Requirement EPC.5.3.3 - The routing service being used - Routing Service Identifier			
EPC .5.4	The resource returned for an Endpoint contains a valid URL	FR	 RAA-4038 ²⁰ - Requirement EPC.5.4 - The resource returned for an Endpoint contains a valid URL			Q: What do we mean by valid? Follows an expected structure or are we testing the link? Probably the former if this is self service

19. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4037?src=confmacro>

20. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4038?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .5.5	The definition (EPC.5.2) and method (EPC5.3) of a desired service must be seen as a unique a destination.	FR	☒ RAA-4039 ²¹ - Requirement EPC.5.5 - The definition (EPC.5.2) and method (EPC5.3) of a desired service must be seen as a unique a destination. IDENTIFIED			

21. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4039?src=confmacro>

Requirement ID	Description	Type (FR / NFR)	Jira Link	Also satisfies	Notes	Comments / Questions
EPC.5.6	The definition of a service must resolve to a destination if present.	FR	 RAA-4040 ²² - Requirement EPC.5.6 - The definition of a service must resolve to a destination if present.			Q: Feels more like a requirement for the re-platformed proxy service rather than the EPC
EPC.5.7	The solution must be able to retrieve endpoint details for any service.	FR	 RAA-4041 ²³ - Requirement EPC.5.7 - The solution must be able to retrieve endpoint details for any service.			

22. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4040?src=confmacro>

23. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4041?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .6	Endpoint information Must be secure.	FR	☒ RAA-4042²⁴ - Requirement EPC.6 - Endpoi nt inform ation Must be secure. IDENTI FIED			
EPC .6.1	The solution must only allow access to the Endpoint information securely.	FR	☒ RAA-4043²⁵ - Requirement EPC.6.1 - The solution must only allow access to the Endpoi nt inform ation securel y. IDENTI FIED			

24. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4042?src=confmacro>

25. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4043?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .6.2	The solution must only allow Authenticated applications to access endpoint information.	FR	 RAA-4044 ²⁶ - Requirement EPC.6.2 - The solution must only allow Authenticated applications to access endpoint information. IDENTIFIED			

26. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4044?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .6.3	As a BaRS implementor the solution must only allow authorised people to access their respective organisation(s) information.	FR	 RAA-4045 ²⁷ - Requirement EPC.6.3 - As a BaRS implementor the solution must only allow authorised people to access their respective organisation(s) information.			

27. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4045?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .7	Endpoint information Must be robust.	FR	☒ RAA-4046 ²⁸ - Requirement EPC.7 - Endpoi nt inform ation Must be robust IDENTI FIED			

28. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4046?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .7.1	The solution must be able to store and retrieve information describing an Organisation, its services and endpoint(s) needed to communicate with them via BaRS.	FR	 RAA-4047 ²⁹ - Requirement EPC.7.1 - The solution must be able to store and retrieve information describing an Organisation, its services and endpoint(s) needed to communicate with them via BaRS.			

29. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4047?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A ls o S a t i s f i e s	Notes	Comments / Questions
EPC .7.2	As a BaRS implementor I must be able to maintain/change information describing the Organisations, Healthcare Services and its Endpoints I am authorised to, easily	FR	 RAA-4048 ³⁰ - Requirement EPC.7. 2 - maintain/ change inform ation describ ing the Organis ations IDENTI FIED			

30. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4048?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .7.3	Any Business Object identifiers must be supported and the solution must be agnostic of those identifiers, for example DoS id or ODS code	FR	 RAA-4049 ³¹ - Requirement EPC.7.3 - Any Business Object identifiers must be supported and the solution must be agnostic of those identifiers, for example DoS id or ODS code IDENTIFIED			Organisation Data Service

31. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4049?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .7.4	The solution must be inline with the BaRS standard for Error Handling.	FR	 RAA-4050 ³² - Requirement EPC.7.4 - The solution must be inline with the BaRS standard for Error Handling.			

32. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4050?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .7.5	The solution must have human readable names for Services, Organisations and Service Discovery items.	FR	 RAA-4051 ³³ - Requirement EPC.7.5 - The solution must have human readable names for Services, Organisations and Service Discovery items.			

33. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4051?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .7.6	The solution must be able to define a timeframe for when an endpoint is available.	FR	 RAA-4052 ³⁴ - Requirement EPC.7.6 - The solution must be able to define a timeframe for when an endpoint is available. IDENTIFIED			
EPC .7.7	The solution must be able to support any kind of Endpoint.	FR	 RAA-4053 ³⁵ - Requirement EPC.7.7 - The solution must be able to support any kind of Endpoint. IDENTIFIED			Future proofing requirement - Refers to different types of endpoints e.g. BARS, ITK etc.

34. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4052?src=confmacro>

35. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4053?src=confmacro>

Requirement ID	Description	Type (FR / NFR)	Jira Link	Also Satisfies	Notes	Comments / Questions
EPC.8	All exposed API must use FHIR R4 to receive and send data.	FR	☒ RAA-4054 ³⁶ - Requirement EPC.8 - All exposed API must use FHIR R4 to receive and send data IDENTIFIED			
EPC.9	The integrity of Endpoint information Must be upheld.	FR	☒ RAA-4055 ³⁷ - Requirement EPC.9 - The integrity of Endpoint information Must be upheld IDENTIFIED			

36. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4054?src=confmacro>

37. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4055?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10	The Endpoint information must be manageable by appropriate actors depending on applicable policy, (different use cases etc).	FR	 RAA-4056 ³⁸ - Requirement EPC.10 - The Endpoint information must be manageable by appropriate actors depending on applicable policy, (different use cases etc). IDENTIFIED			

38. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4056?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 1	Endpoint Information must be accessible by different types implementors, including 3rd parties. e.g. Suppliers, DoS leads, Service provider, BaRS Team.	FR	 RAA-4057 ³⁹ - Requirement EPC.10.1 - Endpoint Information must be accessible by different types implementors, including 3rd parties. e.g. Suppliers, DoS leads, Service provider, BaRS Team.			

39. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4057?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 2	The solution must be transparent to the consumer, direct access to the data must not be possible.	FR	☒ RAA-4058⁴⁰ - Requirement EPC.10.2 - The solution must be transparent to the consumer, direct access to the data must not be possible.			

40. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4058?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 3	The solution must have the ability to view, Add, modify or remove a services endpoint information if I have access to it.	FR	☒ RAA-4059⁴¹ - Requirement EPC.10.3 - The solution must have the ability to view, Add, modify or remove a services endpoint information if I have access to it.			

41. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4059?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 4	The solution must be able to define from a list of known Service Discovery tools, as a source of a services Business identifier. see EPC.5	FR	 RAA-4086 ⁴² - Requirement EPC.10.4 - The solution must be able to define from a list of known Service Discovery tools, as a source of a services Business identifier. see EPC.5 IDENTIFIED			

42. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4086?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 5	The solution must only allow authorised people to access all Endpoint information.	FR	 RAA-4060⁴³ - Requirement EPC.10.5 - The solution must only allow authorised people to access all Endpoint information.			

43. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4060?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 6	The solution must have the ability to Create Endpoint information when requested by an authorised user	FR	 RAA-4062 ⁴⁴ - Requirement EPC.10.6 - The solution must have the ability to Create Endpoint information when requested by an authorised user 			Are we authorising by user or service? Need to figure out the service authenticator layer process

44. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4062?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 6.1	The solution must require all the mandatory data elements defined in the Endpoints fields table	FR	 RAA-4066 ⁴⁵ - Requirement EPC.10 .6.1 - The solution must require all the mandat ory data elemen ts defined in the Endpoi nts fields table IDENTI FIED			

45. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4066?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 6.2	The solution must validate the data elements as defined in the Endpoints fields table	FR	 RAA-4068 ⁴⁶ - Requirement EPC.10.6.2 - The solution must validate the data elements as defined in the Endpoints fields table IDENTIFIED			

46. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4068?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 7	Provide all the data about an Endpoint when requested by an authorised user	FR	☒ RAA-4071 ⁴⁷ - Requirement EPC.10 .7 - Provide all the data about an Endpoi nt when request ed by an authori sed user IDENTI FIED			
EPC .10. 8	Update an Endpoint when requested by an authorised user	FR	☒ RAA-4072 ⁴⁸ - Requirement EPC.10 .8 - Update an Endpoi nt when request ed by an authori sed user IDENTI FIED			

47. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4071?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 9.1	The solution must validate the data elements as defined in the Endpoints fields table	FR	 RAA-4073 ⁴⁹ - Requirement EPC.10.9.1 - The solution must validate the data elements as defined in the Endpoints fields table IDENTIFIED			

48. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4072?src=confmacro>

49. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4073?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .10. 9.2	The user must be a member of one of the maintenance organisations for the endpoint being updated or have administration access	FR	☒ RAA-4074⁵⁰ - Requirement EPC.10.9.2 - The user must be a member of one of the maintenance organisations for the endpoint being updated or have administration access IDENTIFIED			

50. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4074?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .11	Appropriate authorisation for change applied to the Endpoint information for the actors.	FR	 RAA-4075 ⁵¹ - Requirement EPC.11 - Appropriate authorisation for change applied to the Endpoint information for the actors. IDENTIFIED			

51. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4075?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .12	The Solution audits all information access attempts and actions, retaining all pertinent data therein	FR	 RAA-4076 ⁵² - Requirement EPC.12 - The Solution audits all information access attempts and actions, retaining all pertinent data therein IDENTIFIED			

52. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4076?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .13	An accompanying application with a User Interface should be included for internal administration of the Endpoint information.	FR	 RAA-4077 ⁵³ - Requirement EPC.13 - An accompanying application with a User Interface should be included for internal administration of the Endpoint information. REJECTED			Check with Sorcha - Should have been removed

53. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4077?src=confmacro>

Re qui re me nt ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .14	The solution meets a service level agreement that includes a 99.9% uptime guarantee and a high Mean Time Between Failures (MTBF) The solution has adequate disaster recovery and fault tolerance enabling continued operation in the event of a failure	N FR	 RAA-4078⁵⁴ - Requirement EPC.14 - The solution meets a service level agreement that includes a 99.9% uptime IDENTIFIED			
EPC .15	Loss of information must be recoverable	N FR	 RAA-4079⁵⁵ - Requirement EPC.15 - Loss of information must be recoverable IDENTIFIED			

54. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4078?src=confmacro>

55. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4079?src=confmacro>

Re qui re ment ID	Description	T y p e (F R / N F R)	Jira Link	A l s o S a t i s f i e s	Notes	Comments / Questions
EPC .16	The solution must achieve a 90% percentile response time of 0.1 seconds to return an individual resource.	N FR	 RAA-4081 ⁵⁶ - Requirement EPC.16 - The solution must achieve a 90% percentile response time of 0.1 seconds to return an individual resource.			

56. <https://nhsd-jira.digital.nhs.uk/browse/RAA-4081?src=confmacro>